CURRENT SITUATION

Life in Hot and Humid Cities

Work

Without air conditioning you can work only in the cool early-morning hours or at dusk. The first thing I did upon becoming prime minister was to install air conditioners in buildings where the civil service worked. **This was key to public efficiency.** (Lee Kuan Yew)

Travel

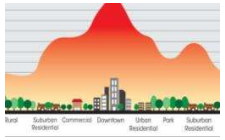
People often use Air conditioned transportation when commuting between work, shop and home.

Shop

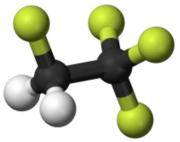
A comfort study showed that subjects who had been exposed more frequently and within a **longer time in AC** rooms were **comfortable at a lower temperature** than those who were regularly being exposed in the naturally ventilated buildings (Tri Harso Karyono).

AC problems

ENERGY CONSUMPTION
the effect of increase in residential energy consumption on the global warming is estimated to be very large. Therefore, it is necessary to improve the standard of living without increasing the energy consumption. (Hiroshi Yoshino)



URBAN HEAT ISLAND
Air-conditioning units funnel heat outside, exacerbating the so-called **"urban heat island"** effect, which makes cities warmer than the countryside. (Chris Bryant)



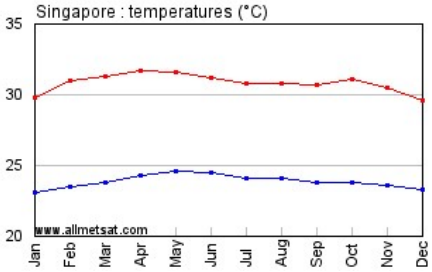
Hydrofluorocarbons (HFCs)
HFCs have high global warming potential, raising concern about their impacts as they become increasingly used as replacements for ozone-depleting substances (ODS), and as economic growth spurs demand for new equipment, especially in the refrigeration/AC sector. (Significant New Alternatives Policy (SNAP))

CURRENT SITUATION

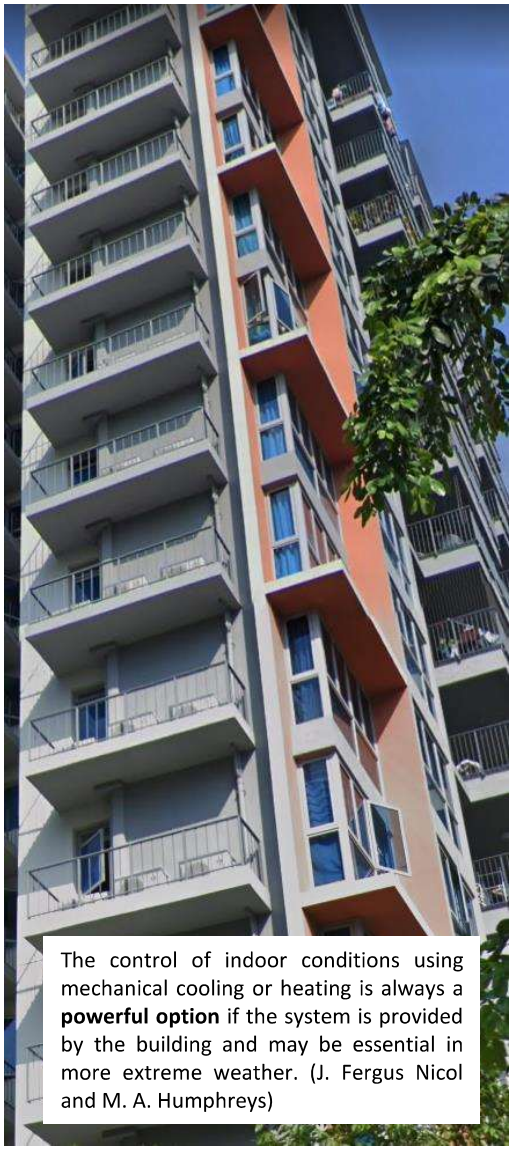
In the hot and humid cities, the **majority of households open their windows during daytime** regardless of AC ownership, while using AC(s) during night-time if they own it (Tetsu Kubota)

In the case of Southeast Asia, people tend to install air-conditioners in their bedrooms rather than in the living room and, as a result, **use them for a long time (approximately 7–8 hours/day) during sleep** (Tetsu Kubota)

No low-income households had ACs. On the other hand, 18% of the middle-income houses and **81% of the high-income houses had ACs.** (Tomoko Uno et al)



The **average temperature settings** were 21.6C (Surabaya) and22.0C (Kuala Lumpur). These values were rather low for ambient temperature setting for sleeping compared with the average value in Japan, which is approximately25.8C. (Tomoko Uno et al)

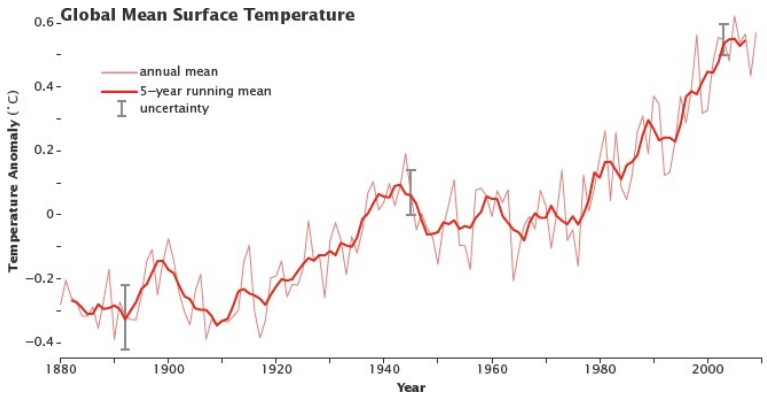


The control of indoor conditions using mechanical cooling or heating is always a **powerful option** if the system is provided by the building and may be essential in more extreme weather. (J. Fergus Nicol and M. A. Humphreys)

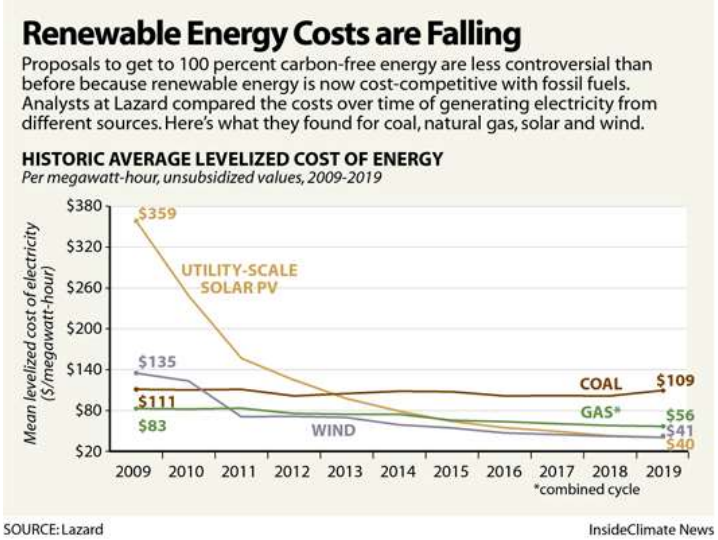
FUTURE TREND

By letting people in overheated climates concentrate on their work and get a good night's sleep, air conditioning has played a big part in driving global prosperity and happiness over the past few decades – and that revolution has still barely begun. (Chris Bryant, Bloomberg)

Global temperature

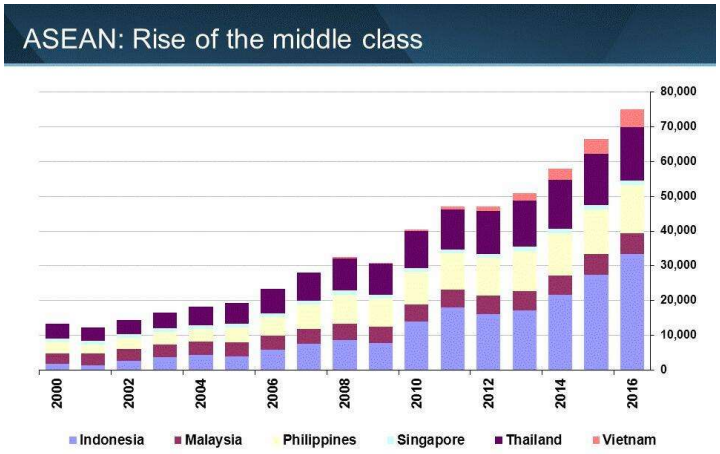


Energy Cost



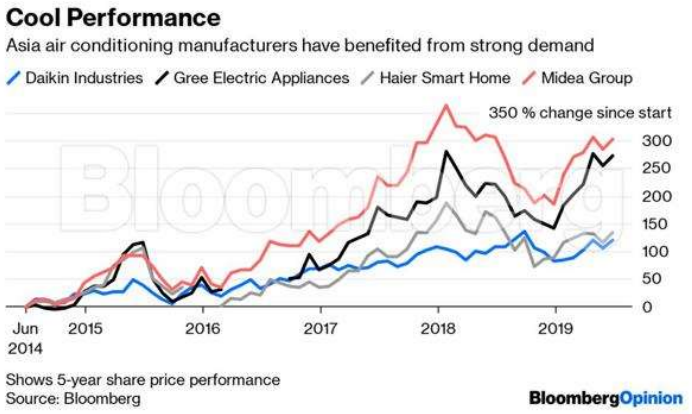
People would use AC if they can afford it

Middle Class



No of households earning >US$10K a year. Source: EIU.

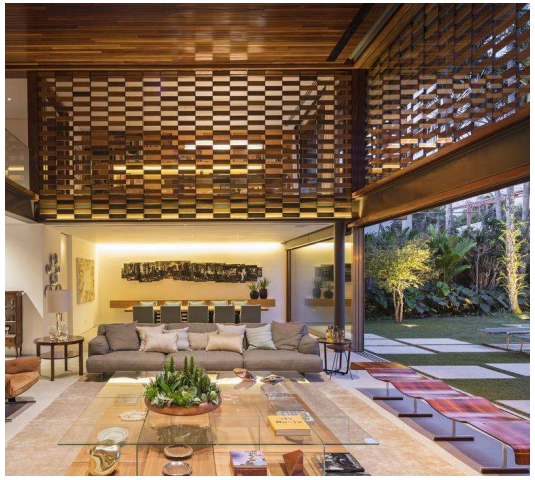
AC Demand



RESEARCH GAP

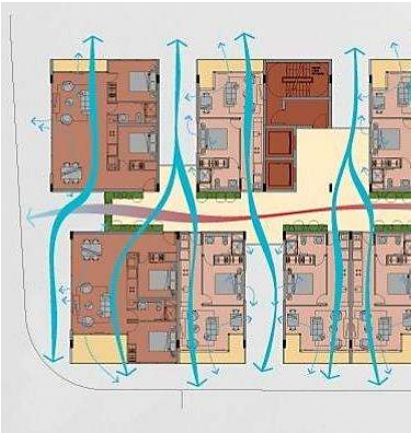
Traditional / High End Housing

Natural ventilation has been the most elementary practice by tradition for cooling in hot-humid climates. (Doris Hooi Chyee Toe)



Researchers have previously considered strategies such as **solar shading, natural ventilation, and façade design**. (Tomoko Uno)

Urban / Middle income housing



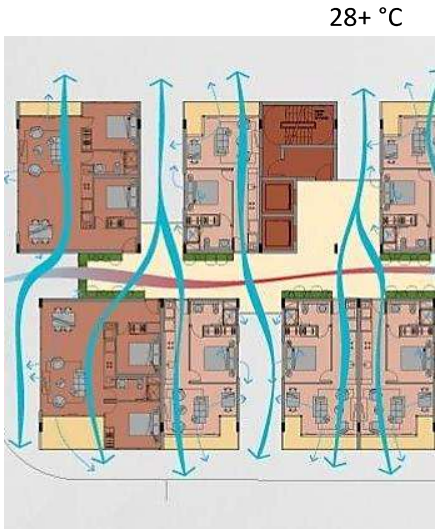
It was determined that residents of houses without AC lived under relatively hot conditions both day and night.

N**atural ventilation** is dependent on outside wind and other conditions and is **not always available.** Instead of relying on the airflow of natural ventilation, people have started to use AC.

Therefore, we should consider the combination of AC use and natural ventilation as a means of effectively saving energy. (Tomoko Uno)

RESEARCH GAP

This study hopes to evaluate the potential improvement of energy efficiency and Internal Air Quality by building houses appropriately designed for AC use.



Energy Consumption

The Energy consumption issue is being addressed by using more renewable energy at the power plant scale.

Hydrofluorocarbons (HFCs)

The HFCs issue is being addressed on the Air Condition unit equipment scale.

URBAN HEAT ISLAND

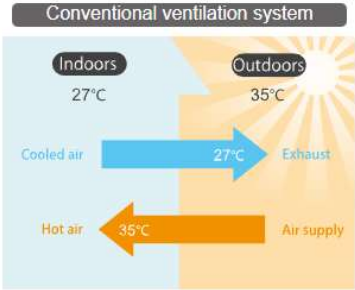
Addressing urban heat island is more effectively done from a building complex or city scale.

COOLING EFFICIENCY

In order to reduce energy use for cooling, it is necessary to understand thermal conditions and the use of air conditioners, together with the identification of **suitable housing designs for cooling.** (Tomoko Uno et al)

PROBLEM STATEMENT

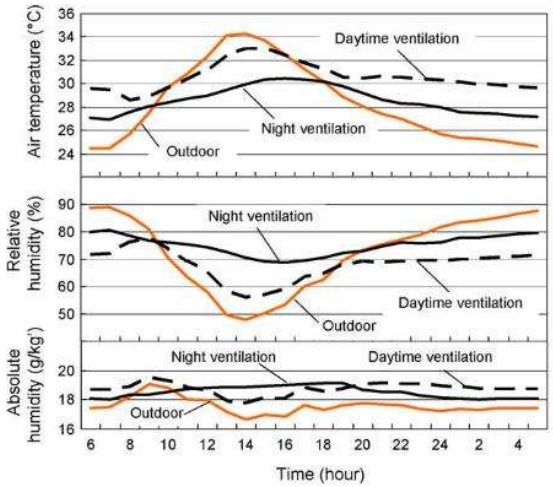
Conventional Hot-Humid Construction: Inefficient use of cooling



Window-opening behavior has become a specific concern when it comes to energy performance of houses. **Windows perform a rather complicated multipurpose function** in houses, and astute occupants need to take an active role in operating them. **Windows play a significant role in heating and cooling** costs (Isabelina Nahmens).

93% of such respondents setting it (AC) under 25C. The temperature settings are lower than the realized conditions in the house. The desired condition may not be achieved because of the insufficient capacity of the AC or **the suboptimal thermal construction of the house.** Since the houses had large openings and were **not airtight**, room temperature is greatly influenced by outdoor conditions. Therefore, even when an AC was used, it may not have maintained the desired room temperature. (Tomoko Uno Et Al)

the room temperature is cooler than the outside air temperature during the day, and it is **hotter than the outside air at night** when the occupant wants to sleep. Therefore, many people come to use the air conditioners to fall asleep. It can be said that **the heat storage performance of concrete and block** does not fit the today's lifestyle and not work effectively. (Hiroto Takaguchi)

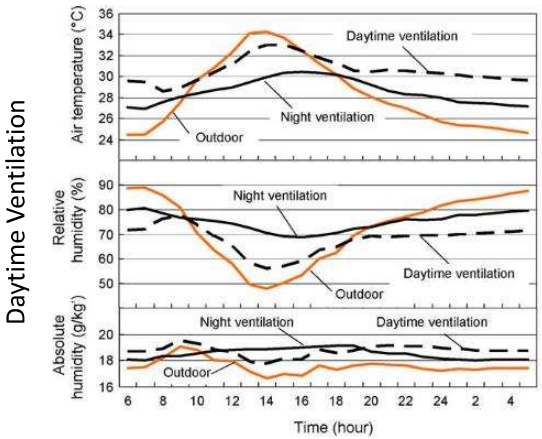


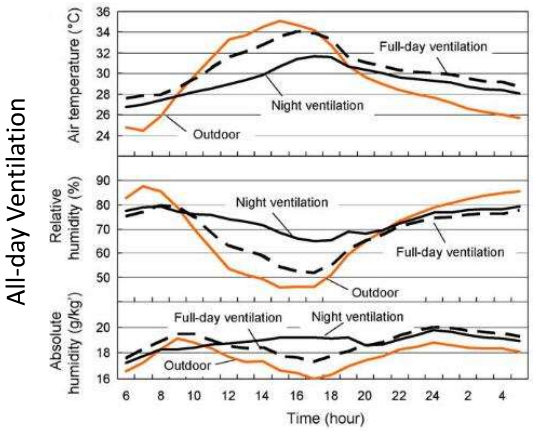
Improvements in the **airtightness of air-conditioned rooms are effective in reducing energy consumption.** This is because such improvements reduce the heat and moisture loads that are due to ventilation from the outdoors. (Tomoko Uno et al)

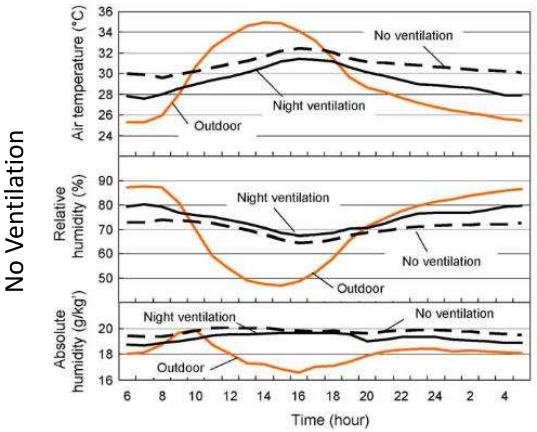
RESEARCH AIM

EVALUATING BUILDING ENVELOPE AND VENTILATION METHODS FOR AIR-CONDITIONED RESIDENCES IN HOT AND HUMID CITIES

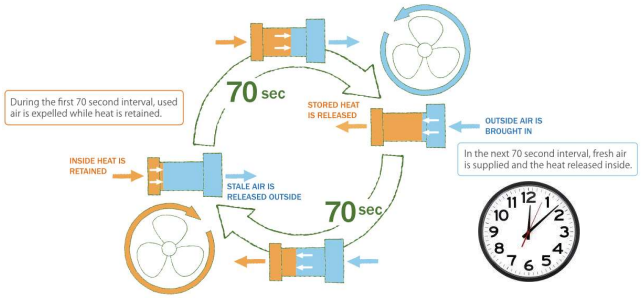
How to effectively use outdoor air to improve Indoor Air Quality?



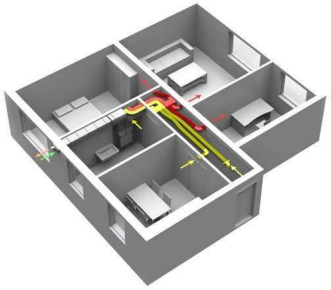




Designing for Thermal Isolation or for Thermal Mass?



Thermal Isolation

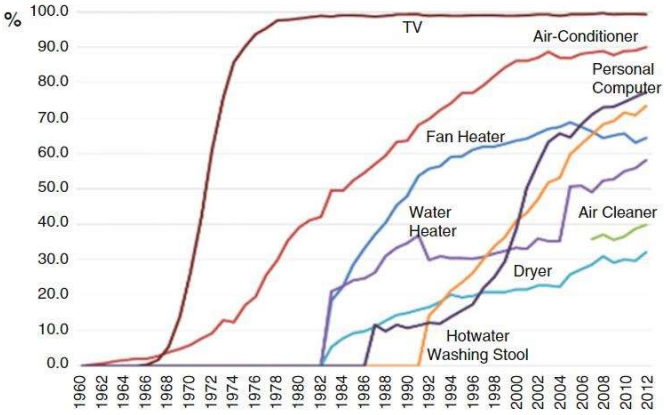


Thermal Mass

RESEARCH QUESTION

What are the alternatives?

Construction techniques used in cooler climate regions: lessons and reference



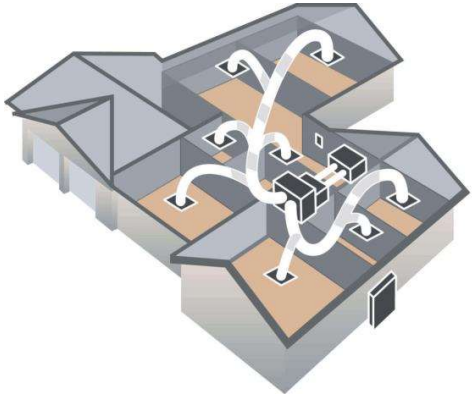
Penetration rate for home appliance in Japan

The air tightness techniques are only seen in recent years because the popularization of air conditioning turned houses into a closed environment.

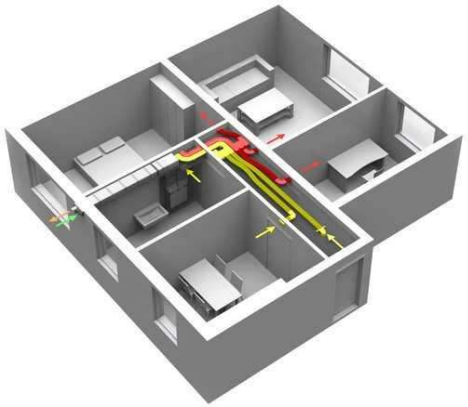
The use of heating and cooling equipment quickly spread to many homes in Japan. By using these appliances to control the indoor environment, **homes became airtight and at the same time poorly ventilated.**

The airtightness in houses, previously mentioned, was found to cause sick house syndrome. (Miwako Nakamura)

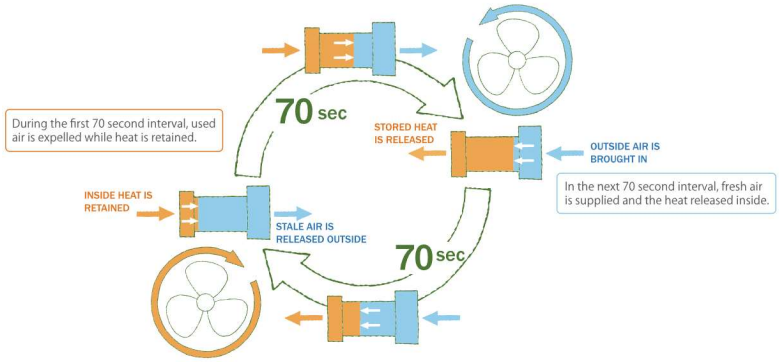
Ventilation Methods



Centralized Fresh Air AC $$$



Ventilation duct $$



Ductless Ventilation $